Fundamentals of Machine Learning

Practice Sheet 1 — Linear Algebra, Change of Basis & Linear Regression

Practice set

Attempt every part in full and write out each computation and derivation rather than convincing yourself you could do it. Mixing the numerical and the conceptual parts is deliberate: the arithmetic checks that the machinery is fluent, and the short-answer parts check that you understand what the machinery *means*. Difficulty runs roughly from medium to hard as you go down the sheet.

Q1. Determinants, eigenvalues and eigenvectors

(a) Compute the determinant of $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$ by cofactor expansion.

(b) For $B = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, find both eigenvalues and a corresponding eigenvector for each. Is B diagonalisable? Justify in one line.

(c) Without recomputing anything, state $\det(B)$ and $\operatorname{tr}(B)$ from your eigenvalues in (b). Then explain in two or three sentences why $\det(M) = 0$ is equivalent to M having 0 as an eigenvalue, and what both statements say about the columns of M.

Q2. Span, rank and linear independence

Let $a_1 = [1,\, 1,\, 0]^\top$, $a_2 = [0,\, 1,\, 1]^\top$, $a_3 = [1,\, 2,\, 1]^\top$, and let $M = [\, a_1\; a_2\; a_3\,]$.

(a) Compute $\det(M)$ and hence $\operatorname{rank}(M)$. Do a_1, a_2, a_3 span $\mathbb{R}^3$?

(b) Give a basis for $\operatorname{span}\{a_1, a_2, a_3\}$ and describe this span geometrically by finding a single linear equation, with constant coefficients α, β, γ, of the form $\alpha x_1 + \beta x_2 + \gamma x_3 = 0$ that every vector $v = [x_1,\, x_2,\, x_3]^\top$ in the span satisfies (here x_1, x_2, x_3 denote the *components* of a general vector v, not the given vectors a_i).

(c) In two or three sentences, tie together the following four statements for a square matrix: "the columns are linearly dependent", "the determinant is zero", "the columns fail to span the whole space", and "the matrix is not invertible".

Q3. Row space, column space and null space

Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$.

(a) Give a basis for each of the four fundamental subspaces of A: the column space $\mathcal{C}(A)$, the row space, the null space $\mathcal{N}(A)$, and the left null space $\mathcal{N}(A^\top)$. State the dimension of each. (Note: one of these subspaces is the zero subspace — say clearly what its basis and dimension are.)

(b) Determine the exact condition on $b = [b_1,\, b_2,\, b_3]^\top$ under which $Ax = b$ has a solution. Relate your condition to one of the subspaces from (a).

Q4. Change of basis

Let $\mathcal{B} = \{\, b_1 = [1,\, 1]^\top,\; b_2 = [1,\, -1]^\top \,\}$, a basis of $\mathbb{R}^2$, and let $P = [\, b_1\; b_2\,]$ be the matrix whose columns are these basis vectors.

(a) Express $x = [3,\, 1]^\top$ (given in the standard basis) in the basis $\mathcal{B}$; that is, find $[x]_{\mathcal{B}}$.

(b) Write down P^{-1} and confirm that $[x]_{\mathcal{B}} = P^{-1}x$.

Q5. Conditioning and ill-conditioned systems

Let $A = \begin{bmatrix} 1.01 & 0.99 \\ 0.99 & 1.01 \end{bmatrix}$.

(a) Find the eigenvalues of A (it is symmetric, so its singular values equal the absolute values of its eigenvalues) and hence its 2-norm condition number $\operatorname{cond}_2(A) = \sigma_{\max}/\sigma_{\min}$.

(b) In two or three sentences, explain why conditioning matters in practice: what an ill-conditioned A implies for gradient-based optimisation of a quadratic objective $\frac{1}{2}x^\top Ax - b^\top x$, and why forming the normal-equations matrix (see Q6) can make a conditioning problem worse.

Q6. Linear regression in closed form

Consider fitting $\hat{y} = w_0 + w_1 x$ to the three data points $(1, 2)$, $(2, 2)$, $(3, 4)$, using the design matrix and target

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \qquad w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}.$$

(a) Starting from the least-squares objective $J(w) = \|Xw - y\|_2^2$, take the gradient and set it to zero to derive the normal equations $X^\top X\, w = X^\top y$. State the closed-form solution.

(b) Compute $X^\top X$, $X^\top y$, and solve for w explicitly. Then report the three residuals $y^{(i)} - \hat{y}^{(i)}$ and verify that they sum to zero.

(c) The closed-form solution can be written $w = X^+ y$ using the pseudoinverse. State the precise condition on X under which $X^\top X$ is invertible (and the solution is unique). Explain in two or three sentences what breaks when that condition fails, and how adding a ridge term — solving $(X^\top X + \lambda I)\, w = X^\top y$ with $\lambda > 0$ — fixes it. Connect your answer to the conditioning discussion in Q5.

Q7. Eigenvalues and Eigenvectors

(a) Find the eigenvalues and a corresponding eigenvector for each, for the upper triangular matrix $C = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. Pick one eigenpair (λ, v) and verify explicitly that $Cv = \lambda v$.

(b) Find the eigenvalues and eigenvectors of the symmetric matrix $D = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, and check that its two eigenvectors are orthogonal (their dot product is zero). State in one line why eigenvectors of a symmetric matrix that correspond to *distinct* eigenvalues are guaranteed to be orthogonal.

(c) For the upper triangular matrix $E = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 5 \end{bmatrix}$, write down its eigenvalues *by inspection* and explain in one sentence why they can be read straight off the diagonal. Then find an eigenvector corresponding to the smallest eigenvalue.

Q8. Closed form on a small dataset

Fit a line $\hat{y} = w_0 + w_1 x$ by least squares to the four points $(0, 1)$, $(1, 3)$, $(2, 5)$, $(3, 8)$, so that

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 8 \end{bmatrix}.$$

(a) Compute $X^\top X$ (a 2×2 matrix) and $X^\top y$.

(b) Invert the 2×2 matrix $X^\top X$ and use $w = (X^\top X)^{-1} X^\top y$ to solve for w_0 and w_1. Write down the fitted line. *(Hint: for* $\begin{bmatrix} a & b \\ b & d \end{bmatrix}$ *the inverse is* $\frac{1}{ad - b^2} \begin{bmatrix} d & -b \\ -b & a \end{bmatrix}$*.)*

(c) Compute the four residuals $y^{(i)} - \hat{y}^{(i)}$ and check that they sum to zero. Is the fit exact? Explain in one sentence what it would take for a straight line to pass through all four points exactly.